Weather Adaptive Autonomous Driving: Methodology (Phase 3 draft)

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3 Methodology

3.1 Baseline: per-weather conformal threshold (Phase 1)

Our baseline is the first step of SeqCRC [1]: conformal risk control (CRC) [2] on the confidence threshold of a frozen detector, with an image-level loss $L_i(\tau)$, the fraction of ground-truth objects in calibration image i with no matched detection scoring at least τ. Calibrated separately in each weather bin b with n calibration images, the threshold is

$$\hat{\tau}(b) \;=\; \max\Big\{\tau \;:\; \tfrac{n}{n+1}\,\hat{R}(\tau;b) + \tfrac{1}{n+1} \le \alpha\Big\}, \qquad \hat{R}(\tau;b) = \tfrac{1}{n}\textstyle\sum_{i=1}^{n} L_i(\tau), \tag{1}$$

which bounds the expected miss fraction of an exchangeable test image from the same bin by α. In Phase 1 we reproduced SeqCRC with its official code (the guarantee held in 5 of 5 configurations on COCO) and ran it on DAWN [3]. Our harness implements Eq. (1) per weather bin on YOLOv8x [4]. With $\alpha = 0.2$ on DAWN, the per-bin threshold keeps the held-out miss fraction at 0.188–0.194 in every bin, at 1.6–6.2 false positives per image (FPPI), while one global threshold misses 0.251 in fog (above α in 98% of 200 splits). On BDD100K [5], a threshold calibrated on clear days misses 0.28–0.36 at night; per-bin thresholds restore the budget at 43–117 FPPI.

Limitation (the gap). The baseline fixes the miss budget and lets the false alarms be whatever results: at night it buys $\alpha = 0.2$ with up to 117 false alarms per image, and there is no way to ask for a different balance. It applies one score cut to every box, it needs the weather bin at run time (our camera-only weather estimators did not isolate fog), and its guarantee assumes exchangeable data within each bin.

3.2 Proposed method: a calibrated trade-off dial

Core idea. Instead of one confidence threshold per weather bin, every detection receives a calibrated probability $p(d)$ that it is a real object, computed from the frozen detector's own outputs, and is kept if and only if $p(d) \ge 1/(1+\lambda)$. The single user-facing parameter λ, the number of false alarms accepted to avoid one missed object, sets the trade-off for the use case. An F-score weight β (and, in Phase 4, a CRC miss budget α) maps onto the same dial.

Table 1 lists what changes relative to the baseline. The detector, data, matching rule and evaluation protocol stay the same; only the decision of which boxes to keep is replaced.

Table 1: What our method keeps, replaces and adds relative to the Phase-1 baseline.

	Baseline (SeqCRC step 1 per weather bin)	Ours (trade-off dial)
Detector	YOLOv8x, frozen COCO weights, 1280 px, score floor 0.001	unchanged
Decision rule	one score threshold per weather bin, $s \geq \hat{\tau}(b)$	per detection, $p(d) \geq 1/(1+\lambda)$; simple option: one threshold $s \geq \tau(\lambda)$
Trade-off	fixed miss budget α; false alarms unpriced	explicit input λ (or β); α planned as a floor via CRC over λ
Run-time inputs	score and the weather bin	score, box context and the detector's per-box outputs (no weather label)
Fitting	calibration half of each bin	once, on 21,831 labelled BDD100K train frames
Added	—	per-box feature tap; calibrated model; $\beta \to \lambda$ map; label-free estimate of false alarms

3.3 Architecture

Figure 1 shows the full pipeline. Online, YOLOv8x runs once per camera frame and returns candidate boxes $D(x)$ with class and score s. We also keep the detector's per-box outputs $h(d)$ at the anchor each box came from: the 80 class probabilities, the four distribution-focal-loss (DFL) distributions over the box sides [6], the neck feature vector, and the pyramid level. A feature builder forms $\varphi(d)$: 20 context features (score, class, size and position, rank, overlap with stronger boxes, crowding) plus 48 per-box features (six class probabilities, strongest other class, top-two margin, sum and entropy of the probabilities, DFL entropy per side and mean, level, and 32 principal components of the neck feature). A gradient-boosted model maps $\varphi(d)$ to $p(d)$, and the dial keeps or drops each box. Offline, the same frozen detector is run once over BDD100K train, boxes are labelled by matching, and the dial is fitted, checked and then graded on data never used for fitting.

3.4 Algorithm

Cost. For a keep rule k over the detections of image x, let $M_k(x)$ be the number of ground-truth objects left without a kept matched box and $F_k(x)$ the number of kept boxes that match no object. The dial setting $\lambda > 0$ defines the cost

$$C_\lambda(k) \;=\; \sum_x \big[\lambda\, M_k(x) + F_k(x)\big]. \tag{2}$$

Option A: one threshold. $k(d) = \mathbf{1}[s(d) \geq \tau(\lambda)]$ with

$$\tau(\lambda) \;=\; \arg\min_\tau \sum_{x \in \mathcal{D}_{\text{train}}} \big[\lambda\, M_\tau(x) + F_\tau(x)\big]. \tag{3}$$

Greedy non-maximum suppression and greedy matching are both score-ordered, so filtering one detector run at τ equals running the detector at τ. Equation (3) is therefore solved exactly from cumulative counts: the minimisers for all λ are the corners of the lower convex hull of (real kept, false kept), stored as 258 switch points.

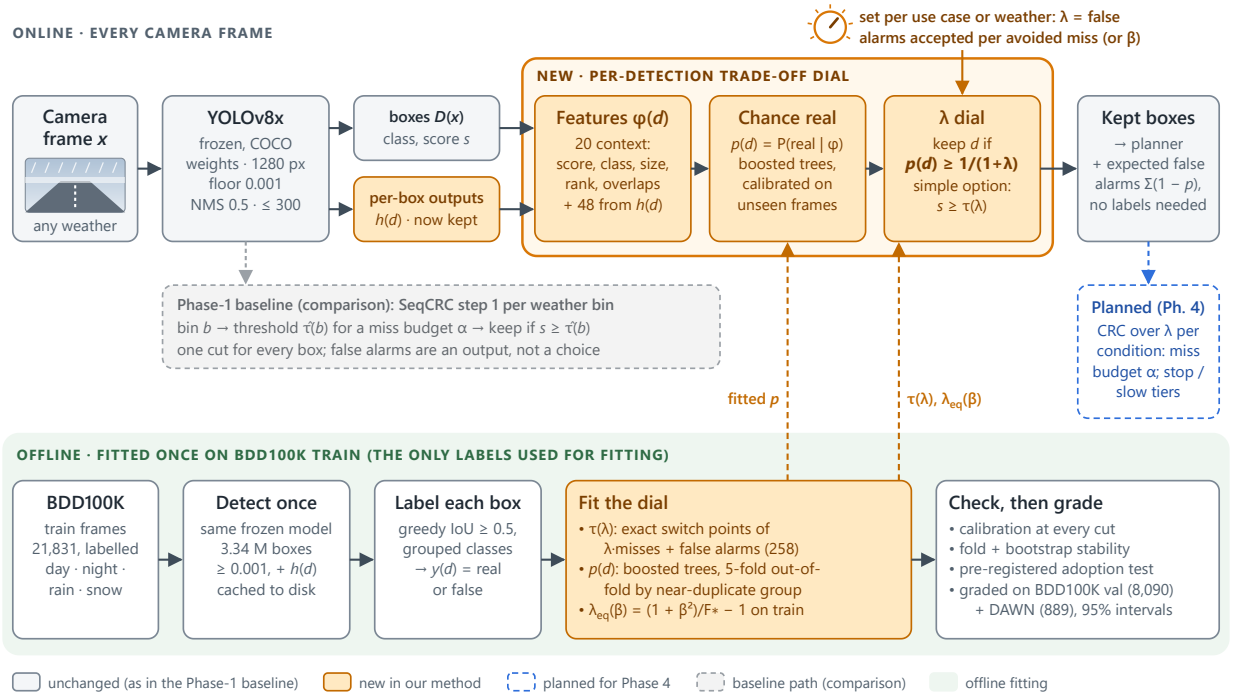


Figure 1: Architecture. New components are orange, unchanged ones grey; the dashed grey lane is the Phase-1 baseline, kept for comparison, and the dashed blue box is planned for Phase 4. Fitting (green band) uses only BDD100K train; grading uses BDD100K val and DAWN.

Option B: per detection. Keeping a box with probability p of being real costs an expected $1-p$ false alarms; dropping it costs an expected λp in misses. The cost-optimal rule is therefore [7]

$$\text{keep } d \iff 1 - p(d) \le \lambda\, p(d) \iff p(d) \ge \frac{1}{1+\lambda}, \tag{4}$$

which is optimal when p is calibrated. Because the kept set grows with λ, the miss count is non-increasing in λ. We estimate $p(d) = \sigma(g(\varphi(d)))$ with gradient-boosted trees [8–10]. They are trained on all train detections by minimising the log-loss $\ell = -\sum_d \left[y_d \log p_d + (1-y_d)\log(1-p_d)\right]$, where $y_d \in \{0,1\}$ comes from greedy IoU-0.5 matching with grouped classes (vehicle, two-wheeler, person). We check calibration with the expected calibration error [11] and with the share of real boxes among detections near each cut $1/(1+\lambda)$.

F-score units. With calibrated probabilities, the F_β-optimal cut is $F_\beta^*/(1+\beta^2)$, where F_β^* is the best achievable F_β [12, 13]. Setting this cut equal to $1/(1+\lambda)$ gives the equivalent dial setting

$$\lambda_{\text{eq}}(\beta) \;=\; \frac{1+\beta^2}{F_\beta^*} - 1, \tag{5}$$

with F_β^* measured on train for each option. For example, F_1 corresponds to $\lambda \approx 1.8$ and F_2 to $\lambda \approx 6.2$ with the per-detection option.

Planned: a miss budget on top of the dial (CRC over λ). Because the image-level miss fraction $L_i(\lambda)$ is non-increasing in λ, the baseline's rule (1) applies to λ itself. Per condition c with

Algorithm 1 Fitting the dial (offline, BDD100K train only)

Require: labelled train frames $\mathcal{D}_{\text{train}}$; frozen detector f

for all $x \in \mathcal{D}_{\text{train}}$ **do**
  $(D(x), h) \leftarrow f(x)$ ▷ floor 0.001, NMS 0.5, ≤ 300 boxes; keep per-box outputs
  $y(d) \leftarrow$ greedy IoU ≥ 0.5 match of $d \in D(x)$ to labels ▷ grouped classes
Option A: sort scores; keep ends of equal-score blocks; cumulative real and false counts
  $\tau(\cdot) \leftarrow$ corners of the lower convex hull ▷ all λ at once
Option B: $\varphi(d) \leftarrow [\text{context}(d, D(x)),\ \text{per-box}(h(d))]$ ▷ PCA fitted on train
  $g \leftarrow$ gradient boosting on $\{(\varphi(d), y(d))\}$ ▷ log-loss
  $p_{\text{oof}} \leftarrow$ 5-fold out-of-fold probabilities, folds by near-duplicate group
for all β **do**
  $\lambda_{\text{eq}}(\beta) \leftarrow (1+\beta^2)/F^*_\beta - 1$ ▷ F-score map, F^*_β on train
check calibration near every cut $1/(1+\lambda)$ and fold stability; adopt only if the pre-registered criteria hold
return $\tau(\cdot)$, g (with PCA), $\lambda_{\text{eq}}(\cdot)$

Algorithm 2 Applying the dial (online, every camera frame)

Require: frame x; setting λ (or β); option $\in$ {A, B}

if β is given **then**
  $\lambda \leftarrow \lambda_{\text{eq}}(\beta)$ ▷ F-score map
if a budget α is certified for condition c **then**
  $\lambda \leftarrow \max(\lambda, \hat{\lambda}_c)$ ▷ planned: CRC over λ
$(D, h) \leftarrow f(x)$ ▷ one detector pass
for all $d \in D$ **do**
  if option A **then**
    keep d iff $s(d) \geq \tau(\lambda)$
  else
    $p(d) \leftarrow \sigma(g(\varphi(d)))$; keep d iff $p(d) \geq 1/(1+\lambda)$ ▷ cost-optimal cut
$\widehat{\text{FA}} \leftarrow \sum_{d \text{ kept}} (1 - p(d))$ ▷ expected false alarms, no labels needed
return kept detections, $\widehat{\text{FA}}$

n_c calibration images,

$$\hat{\lambda}_c \;=\; \inf\Big\{\lambda \;:\; \tfrac{n_c}{n_c+1}\,\hat{R}_c(\lambda) + \tfrac{1}{n_c+1} \leq \alpha\Big\}, \tag{6}$$

and deploying $\max(\lambda_{\text{user}}, \hat{\lambda}_c)$ keeps the CRC guarantee $\mathbb{E}[L_{\text{test}}] \leq \alpha$ under exchangeability within the condition, while never being less cautious than the user asked. The fit is infeasible when α is below the miss fraction at the score floor (11–18% of objects are never detected).

3.5 Design choices

Table 2 gives each design choice, the reason for it, and the alternatives we tried or considered. Most of the alternatives were run in the harness, several under pre-registered criteria.

Table 2: Design choices, the reasons for them, and the alternatives considered (with the evidence).

Choice	Why	Alternatives considered
Keep the detector frozen; decide after it	calibration stays valid; one detector run serves every setting; no retraining per condition	domain adaptation (DA-Detect: 20.41 vs 21.44 AP@50 on real DAWN fog in Phase 1); fine-tuning (invalidates every calibration)
Trade-off as λ, false alarms per avoided miss	use case sets the balance; the optimal rule is explicit (Eq. 4); β and α map onto it	fixed α only (baseline, false alarms unpriced); maximise F_1 (one fixed balance); FPPI cap
Decide per detection	87–91% of a per-image oracle's advantage on BDD100K is chance; the learnable signal is in individual boxes	per-image thresholds from pixels, from the confidence profile, or from YOLOv8x image features (all failed pre-registered tests); box-combination rules; summed-confidence maps
One threshold kept as the simple option	one auditable number per λ; per-weather thresholds give no gain (0.98–1.03× the cost)	per-weather thresholds (the baseline's form)
Gradient-boosted trees on detector outputs	tabular inputs; trains in minutes on CPU; calibrated without post-hoc scaling (expected calibration error 0.11 points on BDD100K val)	recalibrating the score alone (this is Option A); a neural head; retraining YOLOv8x's head
YOLOv8x's own per-box outputs as features	pre-registered gain: cost 0.972× (0.969–0.975) of the context-only model on BDD100K, not worse on DAWN (0.995)	context features only (model 1); a separate CNN on box crops (more compute)
One model for every β	theory: only the cut moves with β; dedicated β-weighted models tie (0.997–1.001×)	one cost-weighted model per β
Fit on BDD100K train; grade on BDD100K val and DAWN	nothing is chosen on graded data; DAWN tests transfer to real fog, rain, sand and snow	calibrating on halves of the test pool (the baseline protocol)
Grouped classes in matching	car/truck confusions are not safety misses	class-aware matching (run 1)

3.6 Experiment plan (Phase 4)

Data and splits. The dial is fitted on 21,831 BDD100K train frames (3.34 M detections), with 5-fold out-of-fold estimates grouped by near-duplicate group. It is graded on BDD100K val (8,090 frames in seven weather-by-time bins) and on DAWN (889 frames with clean labels: fog 256, rain 187, sand 243, snow 203). Neither set is used for any choice. For CRC over λ we use 200 random calibration/test halves per condition, keeping near-duplicate groups together, as in the baseline runs.

Metrics. The main metric is the cost C_λ relative to one threshold, with 95% image-bootstrap intervals [14]. We also report recall, precision and false alarms per image at $\lambda \in \{1, 3, 10, 30, 100\}$; F_β at $\beta \in \{0.5, 1, 2\}$ against no tuning and the best in-sample constant; and calibration (expected calibration error, share real near each cut). For CRC over λ we report the held-out miss fraction against α (share of splits above), its price in false alarms per image, and the smallest certifiable α.

Baselines. (i) No tuning ($\tau = 0.25$, the Ultralytics default). (ii) One global CRC threshold. (iii) The per-weather CRC threshold, Eq. (1) (Phase-1 baseline). (iv) Option A, the one-threshold dial. (v) The context-only per-detection model. As reference bounds that no deployable rule can reach, we also report the best constant threshold chosen in-sample per condition and the per-frame hindsight best.

Ablations. (a) Retraining without each feature group: class probabilities, neck components, DFL entropy, level, context. (b) Per-condition recalibration, which targets the under-stated probabilities in DAWN fog (17% real at the 9.1% cut). (c) Training-set size: 25, 50 and 100% of frames. (d) Detector size and input resolution: YOLOv8x at 640 versus 1280, and YOLOv8s. (e) CRC over λ versus CRC over τ at equal α. Already completed: model 1 versus model 2, per-weather thresholds, and dedicated β models.

Compute. Measured on our laptop GPU (RTX 3050 Ti, 4 GB):

- YOLOv8x at 1280 px takes 0.15–0.17 s per frame;
- extracting per-box outputs for all 30.9k frames takes about 85 minutes;
- fitting the per-detection model takes about 11 minutes on the CPU;
- the CRC splits take minutes on the CPU.

We will use an NRP A100 for the new detector passes in ablation (d) and for the repeated fits in (a)–(c). At an estimated $\geq 10\times$ the laptop's throughput, these take a few GPU-hours in total. The laptop remains a fallback for every step.

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